APEX RETAIL GROUP | PRACTICAL CLASS TASKS

DOCUMENT: CLASS TASKS (SESSION 4)

SCENARIO: E-COMMERCE TRANSACTION DATASET DISCOVERY

INSTRUCTIONS FOR LEARNER: Complete the following four tasks in your Jupyter Notebook during class. Follow the structured workflow: **Data Loading -> Structural Inspection -> Metadata Auditing -> Business Interpretation**. Do not peek at instructor solution keys until completed.

TASK 1: LOADING & STRUCTURAL INSPECTION

Samantha Reed (E-Commerce Manager) provides a batch transaction dataset representing recent web store orders:

```Python
import pandas as pd

raw_orders = {
    "order_id": ["ORD-201", "ORD-202", "ORD-203", "ORD-204", "ORD-205", "ORD-206", "ORD-207", "ORD-208"],
    "date": ["2026-10-10", "2026-10-10", "2026-10-11", "2026-10-11", "2026-10-12", "2026-10-12", "2026-10-13", "2026-10-13"],
    "category": ["Electronics", "Apparel", "Home", "Electronics", "Apparel", "Home", "Electronics", "Apparel"],
    "quantity": [1, 3, 2, 1, 5, 4, 2, 1],
    "revenue": [1200.00, 150.00, 80.00, 899.00, 250.00, 160.00, 1798.00, 75.00],
    "payment_method": ["Credit Card", "PayPal", "Credit Card", "Debit Card", "PayPal", "Credit Card", "Credit Card", "Debit Card"]
}
```

1. Import `pandas` as `pd` and construct a DataFrame named `df_orders` from `raw_orders`.
2. Print the dimensions (`.shape`) of `df_orders`.
3. Display the first 4 rows (`.head(4)`) and the last 2 rows (`.tail(2)`).

TASK 2: METADATA AUDITING & COLUMN SELECTION

1. Run `df_orders.info()` to inspect data types and non-null counts.
2. List all column names using `.columns`.
3. Create a sub-DataFrame named `df_revenue` that extracts only the `order_id`, `category`, and `revenue` columns. Print `df_revenue`.

TASK 3: SUMMARY STATISTICS & ANOMALY DISCOVERY

1. Execute `df_orders.describe()` to generate summary statistics for numerical columns (`quantity`, `revenue`).
2. Identify the mean revenue, minimum revenue, and maximum revenue across orders.
3. Calculate the total revenue generated across all orders using Pandas' `.sum()` method on the `revenue` column.

TASK 4: EXECUTIVE DISCOVERY SUMMARY

Review your outputs from Tasks 1, 2, and 3:

1. How many total orders are in the dataset, and what is the total revenue generated?
2. What is the data type assigned to the `date` column? Is it a date object or a string (`object`)?
3. Draft a 2-3 sentence dataset health summary to Samantha Reed confirming dataset structure and key revenue metrics.